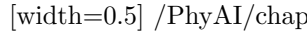
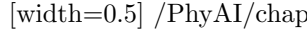


Magnetism in Matter I think I can safely say that nobody understands quantum mechanics. *Richard Feynman*
 A loop inside magnetic field $\vec{B}$ A rectangular loop has size $a \times b$, under the net torque

I define $\vec{n}$ as the normal vector of the loop, and an angle $\theta = (\vec{n}, \vec{B})$, because both torques vectors are in the same direction.
 Both torque vectors in pointing out from the plane of paper The scalar τ_{net} value is:

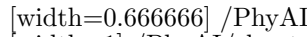
I define a new vector $\vec{\mu}$ quantity, called **magnetic dipole moment**, same direction to normal vector $\vec{n}$, and has magnitude

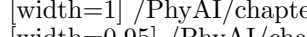
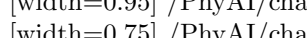
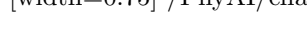
Rewriting the net torque equation yields:

[Magnetic Moment Definition] The **magnetic moment** (or magnetic dipole moment) is a vector quantity that determines t

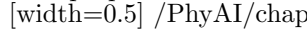
Each angle θ (position), it is characterized by a **potential energy value**. I randomly choose 2 angles θ_i and θ_f :

I assign the potential energy value to angle θ :

From here we have a very important result: a magnetic dipole has its lowest energy when its vector $\vec{\mu}$ is lined up with magn
 Highest and lowest potential energy value To flip an arrow (magnetic moment v
 A counterclockwise current inside a diverging magnetic field

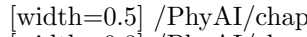
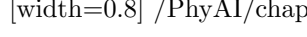
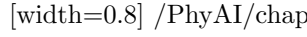
[H]
 A clockwise current inside a diverging magnetic field [H]
 Two loops system tends to be **repelled away** from the north pole Inside Bohr's ato
 The macroscopic view of paramagnetic material An external magnetic field $\vec{B}$ imme

In 1985, Pierre Curie discovered experimentally the relationship:

This proportional relationship seems like very obvious; the higher the temperature, the harder the alignment of magnetic m
 Two-state paramagnet If I denote N as "the number of", the multiplicity and entrop

The internal energy is determined by:

I neglect the constant B in the magnetization formula, and redefine the M quantity as:

Before we look for an analytical solution, let's consider the visual representation for these things. We all know how binomial
 Relationship between $N_{\uparrow}, S, U, M$ If our two-state paramagnet starts moving from sta
 The graph of (U, S) What can we see here? Recall the definition of temperature (D
 (U, T) and (T, M) graphs By sketching our data on graphs, the fact that $M \propto \frac{1}{T}$ st
 $\approx k_B [N \ln N - N_{\uparrow} \ln N_{\uparrow} - (N - N_{\uparrow}) \ln (N - N_{\uparrow})]$ The temperature function is:
$$T = \left(\frac{\partial S}{\partial U} \right)_{N, V, B} = \frac{\partial S}{\partial N_{\uparrow}} \cdot \frac{\partial N_{\uparrow}}{\partial U} = \frac{\partial S}{\partial N_{\uparrow}} \cdot \left(\frac{\partial U}{\partial N_{\uparrow}} \right)^{-1} = k_B \left[\frac{1}{N_{\uparrow}} - \frac{1}{N - N_{\uparrow}} \right]$$

$N_{\downarrow} = \frac{N + U/\mu B}{2}$
 We obtain:

I rearrange some terms here: $\exp\left(\frac{2\mu B}{k_B T}\right) = \frac{N - U/\mu B}{N + U/\mu B}$

$$\Leftrightarrow \frac{N\mu B - U}{N\mu B + U} = \exp\left(\frac{2\mu B}{k_B T}\right)$$

$$\Leftrightarrow N\mu B - U = (N\mu B + U) \exp\left(\frac{2\mu B}{k_B T}\right)$$

$$\Leftrightarrow N\mu B \left[1 - \exp\left(\frac{2\mu B}{k_B T}\right)\right] = U \left[1 + \exp\left(\frac{2\mu B}{k_B T}\right)\right]$$

$$\Leftrightarrow U = N\mu B \left[\frac{1 - \exp(2\mu B/k_B T)}{1 + \exp(2\mu B/k_B T)} \right]$$
 From the definition of $\tanh x$ function: